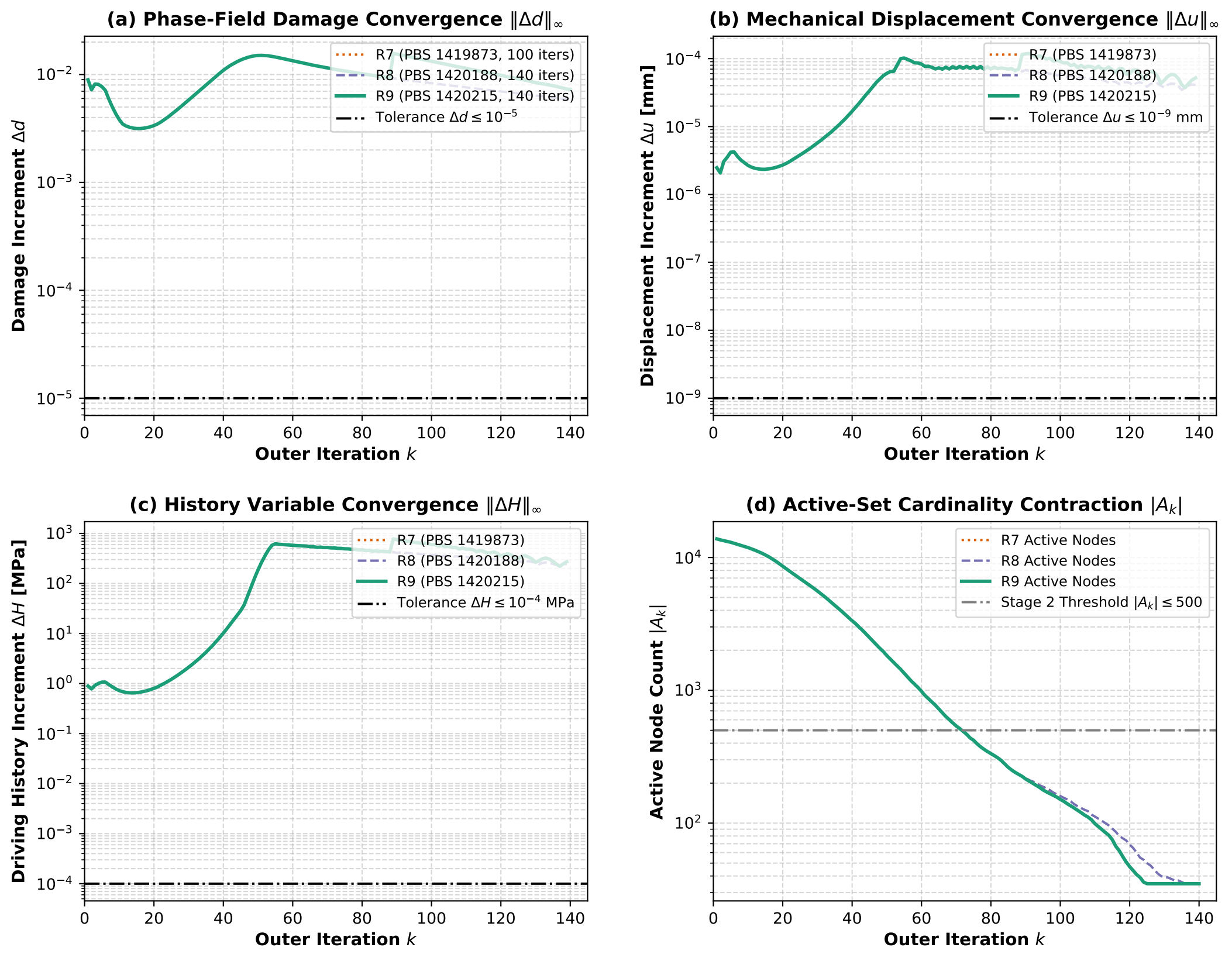
(a) Phase-Field Damage Convergence $\|\Delta d\|_\infty$
Damage Increment Δd
Outer Iteration k
R7 (PBS 1419873, 100 iters)
R8 (PBS 1420188, 140 iters)
R9 (PBS 1420215, 140 iters)
Tolerance $\Delta d \le 10^{-5}$
(b) Mechanical Displacement Convergence $\|\Delta u\|_\infty$
Displacement Increment Δu [mm]
Outer Iteration k
R7 (PBS 1419873)
R8 (PBS 1420188)
R9 (PBS 1420215)
Tolerance $\Delta u \le 10^{-9}$ mm
(c) History Variable Convergence $\|\Delta H\|_\infty$
Driving History Increment ΔH [MPa]
Outer Iteration k
R7 (PBS 1419873)
R8 (PBS 1420188)
R9 (PBS 1420215)
Tolerance $\Delta H \le 10^{-4}$ MPa
(d) Active-Set Cardinality Contraction $|A_k|$
Active Node Count $|A_k|$
Outer Iteration k
R7 Active Nodes
R8 Active Nodes
R9 Active Nodes
Stage 2 Threshold $|A_k| \le 500$